NAME -: ABHISHEK AGARWAL **CLASS ROLL NO -: 01**

UNIVERSITY ROLL NO -: 12515000047 **SECTION -: FB**

ASSIGNMENT – 01

1464. Maximum Product of Two Elements in an Array.

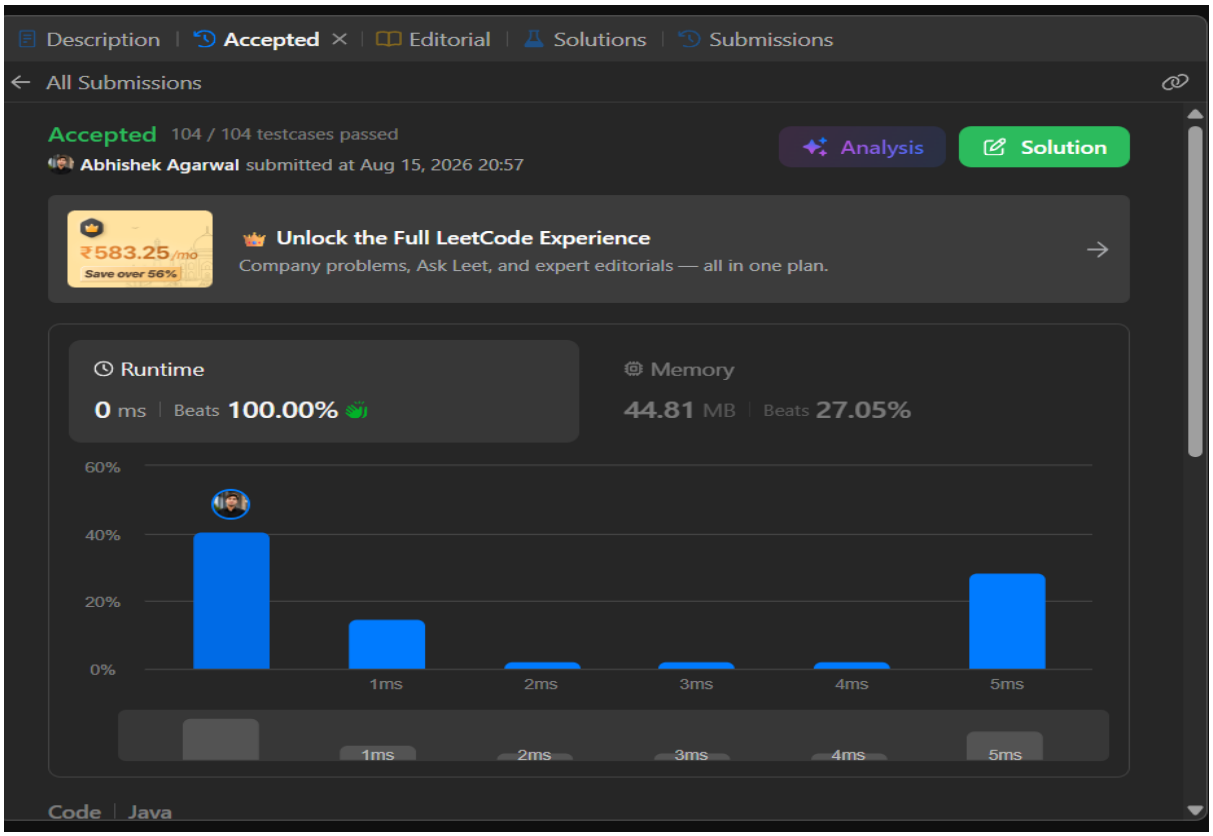
Code | Java

```java
class Solution {
    public int maxProduct(int[] nums) {
        int max1 = 0;
        int max2 = 0;

        for (int num : nums) {
            if (num > max1) {
                max2 = max1;
                max1 = num;
            } else if (num > max2) {
                max2 = num;
            }
        }

        return (max1 - 1) * (max2 - 1);
    }
}
```

View less



162. Find Peak Element.

Code | Java

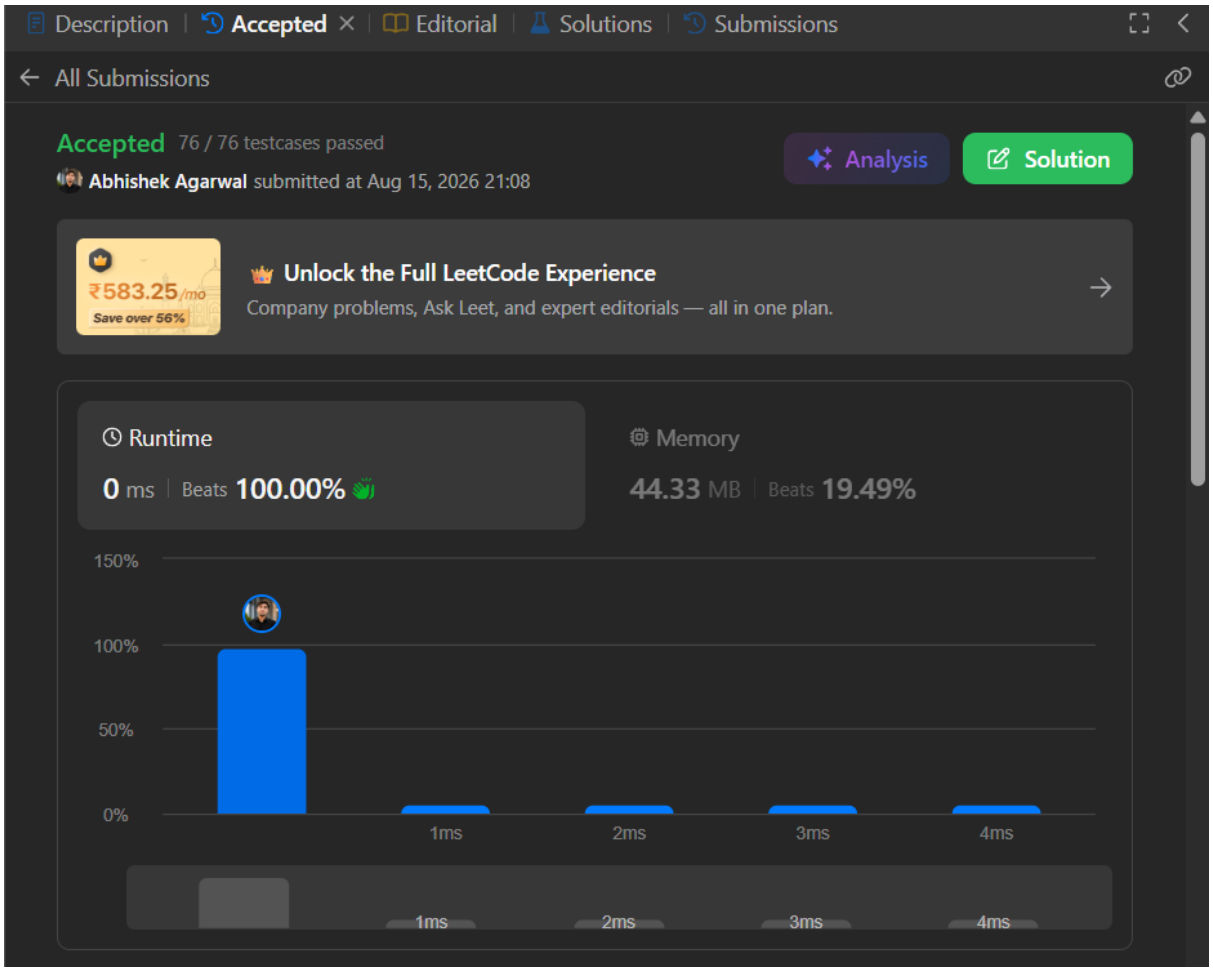
```java
class Solution {
    public int findPeakElement(int[] nums) {
        int left = 0;
        int right = nums.length - 1;

        while (left < right) {
            int mid = left + (right - left) / 2;

            if (nums[mid] > nums[mid + 1]) {
                right = mid;
            } else {
                left = mid + 1;
            }
        }

        return left;
    }
}
```

View less



33. Search in Rotated Sorted Array.

Code | Java

```java
class Solution {
    public int search(int[] nums, int target) {
        int left = 0;
        int right = nums.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (nums[mid] == target)
                return mid;

            // Left half sorted
            if (nums[left] <= nums[mid]) {

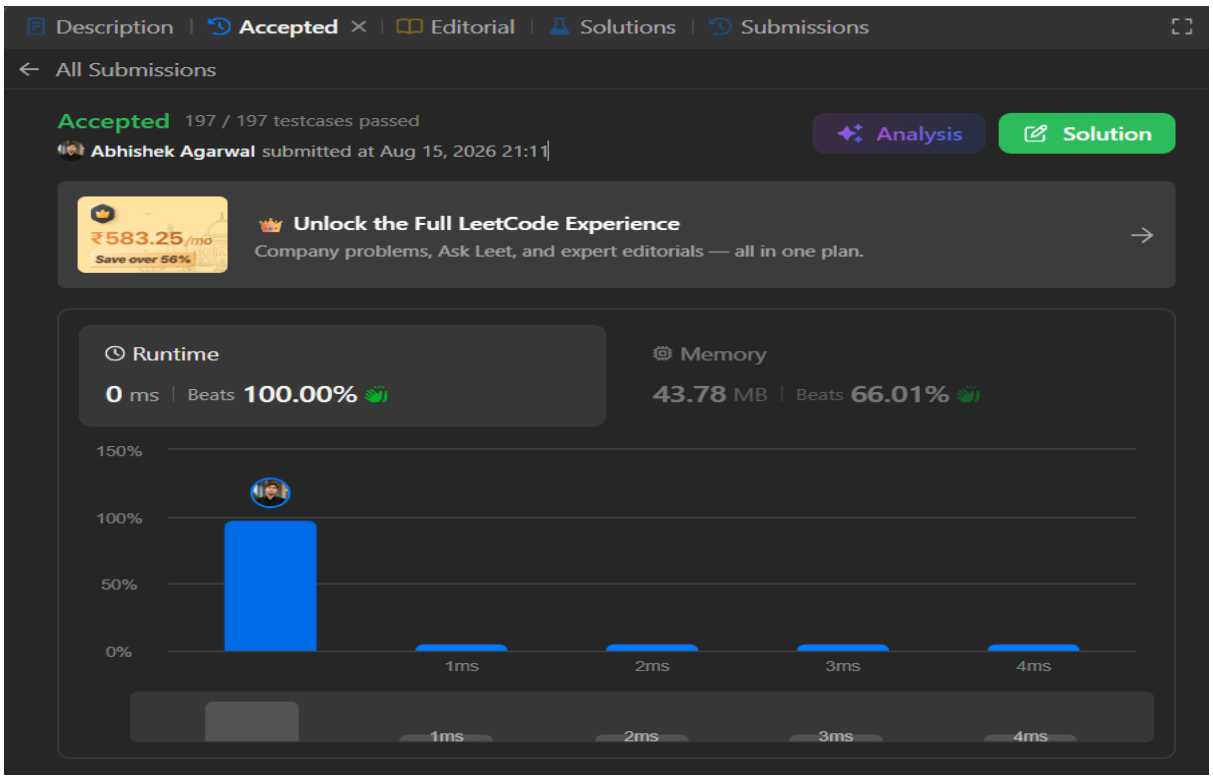
                if (nums[left] <= target && target < nums[mid])
                    right = mid - 1;
                else
                    left = mid + 1;

            // Right half sorted
            } else {

                if (nums[mid] < target && target <= nums[right])
                    left = mid + 1;
                else
                    right = mid - 1;
            }
        }

        return -1;
    }
}
```

View less



3477. Fruits Into Baskets II.

Code | Java

```java
class Solution {
    public int numOfUnplacedFruits(int[] fruits, int[] baskets) {

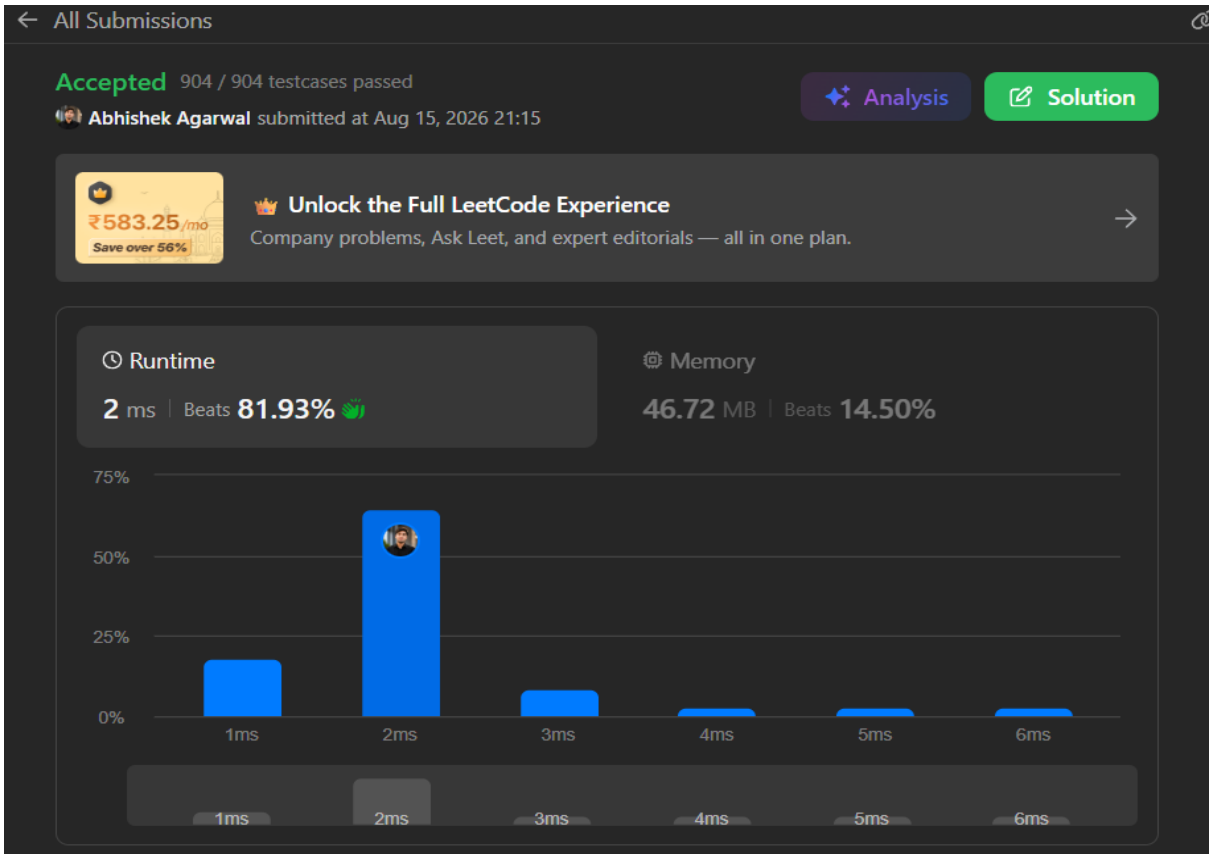
        int n = fruits.length;
        boolean[] used = new boolean[n];
        int unplaced = 0;

        for (int i = 0; i < n; i++) {
            boolean placed = false;

            for (int j = 0; j < n; j++) {
                if (!used[j] && baskets[j] >= fruits[i]) {
                    used[j] = true;
                    placed = true;
                    break;
                }
            }

            if (!placed)
                unplaced++;
        }

        return unplaced;
    }
}
```

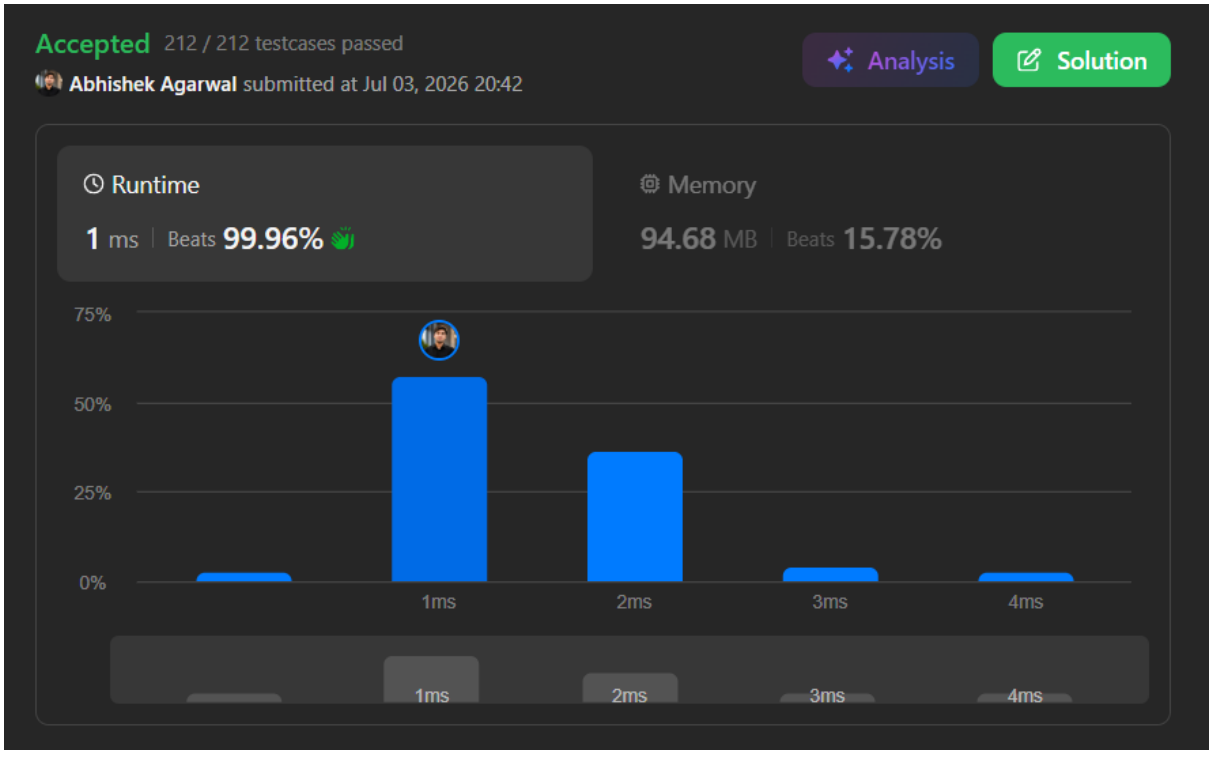


121. Best Time to Buy and Sell Stock.

Code | Java

```java
class Solution {
    public int maxProfit(int[] prices) {
        int maxProfit=0;
        int buy=prices[0];
        for(int i=0;i<prices.length;i++){
            int sell = prices[i];
            if(buy>sell){
                buy=sell;
            }
                int profit=sell-buy;
                if(profit>maxProfit){
                    maxProfit =profit;
                }
        }
         return maxProfit;
    }
}
```

View less



1. Two Sum.

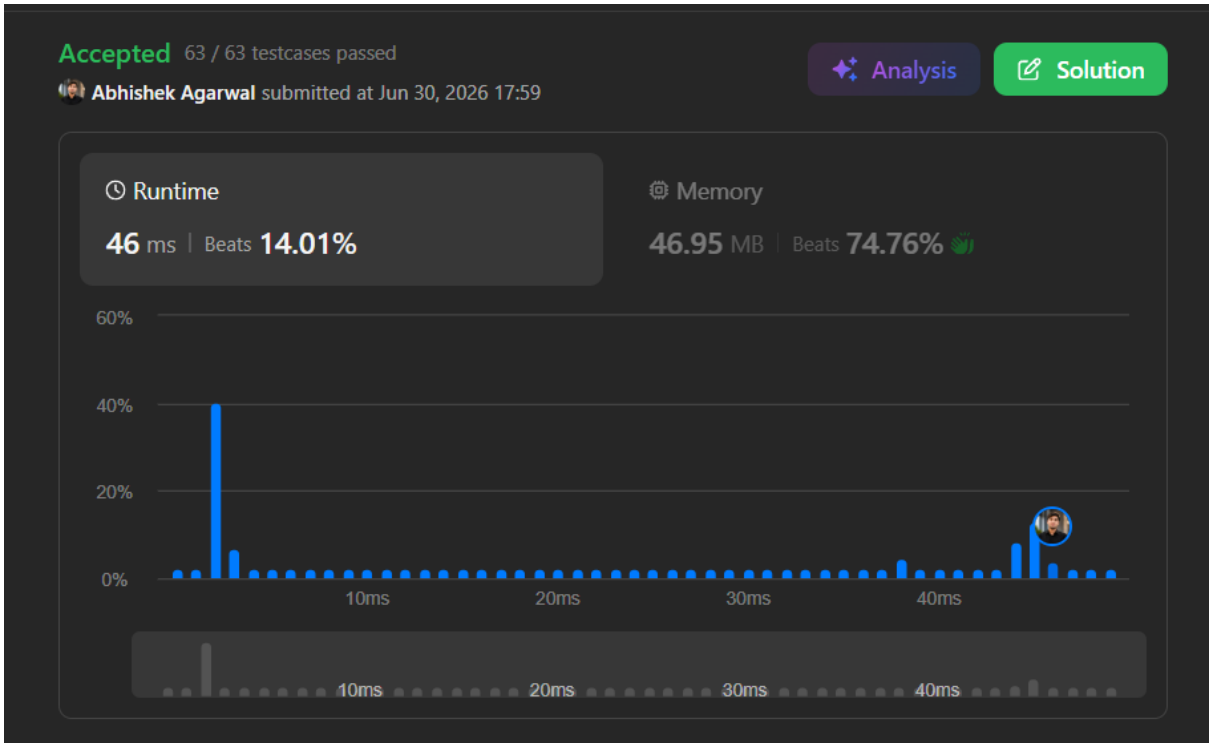
Code | Java

```java
class Solution {
    public int[] twoSum(int[] nums, int target) {

        for (int i = 0; i < nums.length; i++) {
            for (int j = i + 1; j < nums.length; j++) {
                if (nums[i] + nums[j] == target) {
                    return new int[] { i, j };
                }
            }
        }

        return new int[] {};
    }
}
```

View less



1365. How Many Numbers Are Smaller Than the Current Number.

Code | Java

```java
class Solution {
    public int[] smallerNumbersThanCurrent(int[] nums) {

        int[] ans = new int[nums.length];

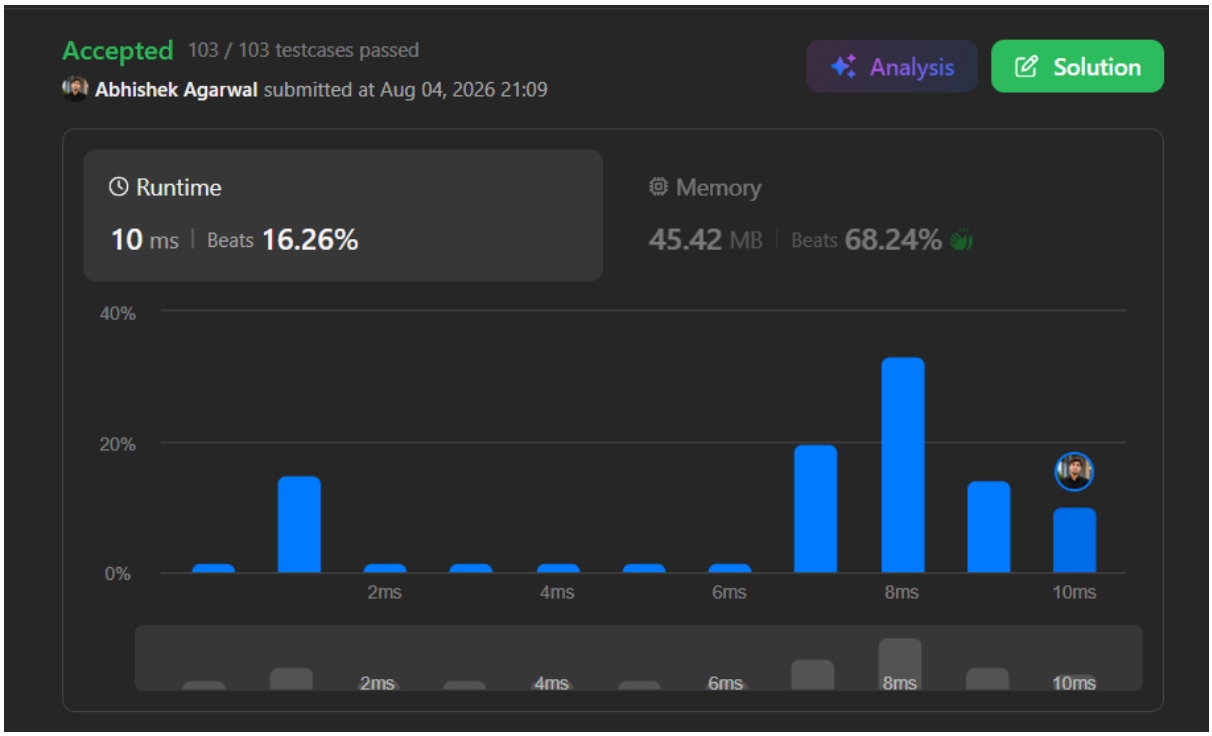
        for (int i = 0; i < nums.length; i++) {
            int count = 0;

            for (int j = 0; j < nums.length; j++) {
                if (nums[j] < nums[i]) {
                    count++;
                }
            }

            ans[i] = count;
        }

        return ans;
    }
}
```

View less



628. Maximum Product of Three Numbers.

Code | Java

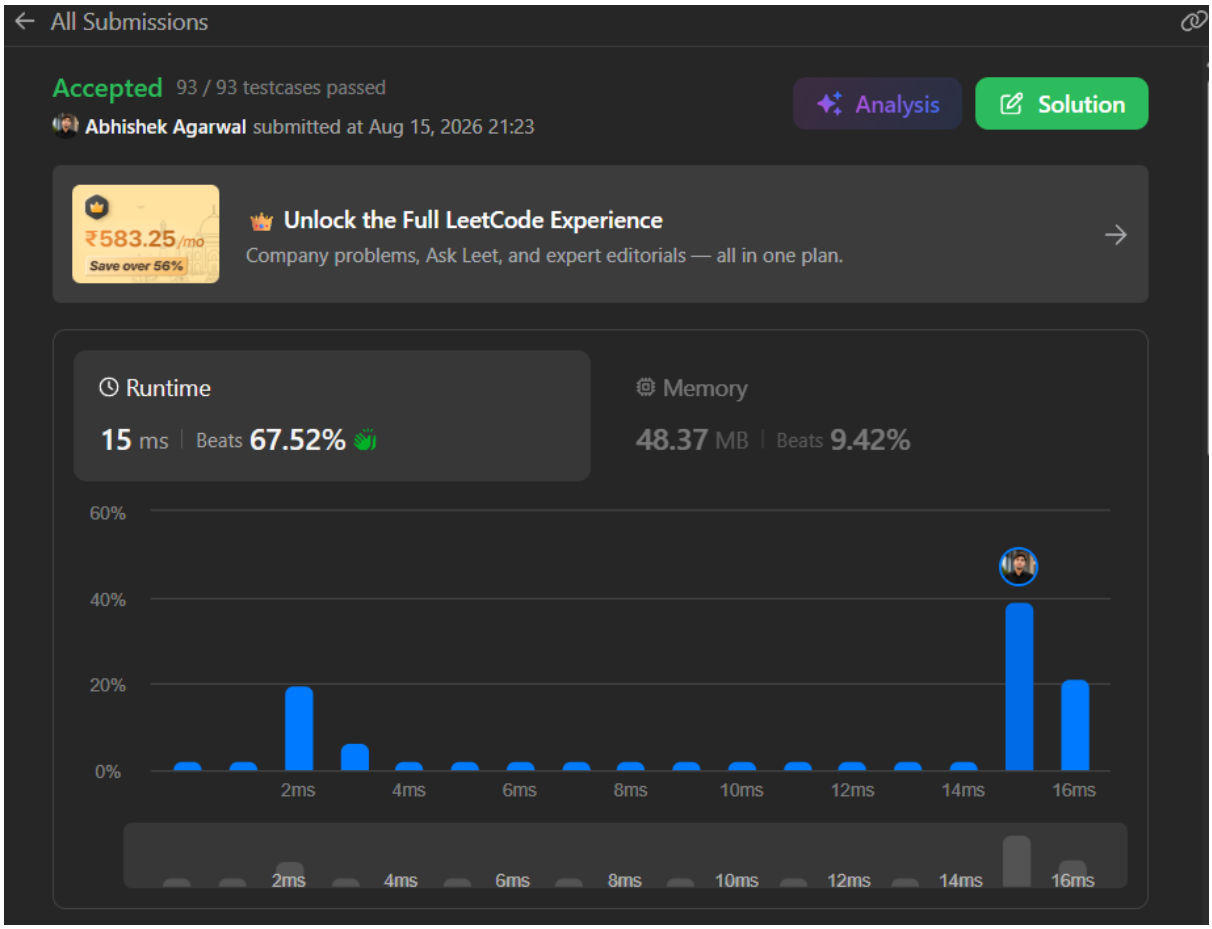
```java
import java.util.Arrays;

class Solution {
    public int maximumProduct(int[] nums) {
        Arrays.sort(nums);

        int n = nums.length;

        return Math.max(
            nums[n - 1] * nums[n - 2] * nums[n - 3],
            nums[0] * nums[1] * nums[n - 1]
        );
    }
}
```

View less



26. Remove Duplicates from Sorted Array.

Code | Java

```java
class Solution {
    public int removeDuplicates(int[] nums) {
        int i=0;
        for(int j=1;j<nums.length;j++){
            if(nums[i]!=nums[j]){
                i++;
                nums[i]=nums[j];
            }
        }
        return i+1;
    }
}
```

View less

